

Research Note: Re-architecting the Low-Level Design (LLD) Practice Loop

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1. Executive Summary

Low-Level Design (LLD) / Object-Oriented Design (OOD) is a core hiring competency for Software Engineers across top technology organizations. While Data Structures and Algorithms (DSA) benefit from mature automated practice ecosystems (e.g., LeetCode, HackerRank, Codeforces), **Low-Level Design practice remains profoundly broken.**

Learners today can read dozens of reference designs for a "Parking Lot" or "Elevator System," but when asked to produce their own design, they face an evaluative vacuum. Because software design inherently allows multiple valid architectures with distinct trade-offs, learners cannot determine whether their classes exhibit high cohesion, whether their abstractions are leaky, or whether their design will withstand evolving requirements.

This research note analyzes the cognitive hurdles of practicing LLD, critically reviews existing industry alternatives, identifies three foundational gaps, and articulates our product direction for a focused, rubric-grounded LLD practice platform.

2. The Learner Problem: Why LLD Practice is Hard

In our investigation of learner workflows and design interview preparation patterns, we identified four interconnected failure modes in the self-practice journey:

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1. Ambiguity of "Multiple Valid Truths" | 2. The "Passive Reading" Trap | 3. Evaluative Black Hole | 4. No Iterative Feedback Loop
Valid designs look like studying diagrams. "Did I violate SRP? Is my code actually coupled?" - No answer improved over 1

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2.1 The Ambiguity of "Multiple Valid Truths"

In DSA, correctness is verified through automated unit tests over boundary input/output pairs. In LLD, two entirely different designs for the same problem can both be correct, depending on which non-functional requirement is prioritized: * A parking lot using a simple List with synchronized methods is optimal for small garages. * An enterprise parking lot delegating to a ParkingStrategy (Strategy Pattern) with tiered spot types and observer-based occupancy monitors is optimal for multi-floor scale.

Without guidance, learners conflate functional completeness with design quality. They assume

that if their pseudocode compiles or "does the job," their Low-Level Design is sound.

2.2 The "Passive Reading" Illusion

Learners disproportionately consume passive resources (blogs, YouTube walkthroughs, GitHub repositories). When reading an expert's pre-baked class diagram, the design choices appear obvious in hindsight. However, during an interview or active practice session, the learner struggles to decompose requirements into distinct domain entities, define interfaces, and assign single responsibilities.

2.3 The Evaluative Black Hole

When a candidate finishes drafting an LLD solution on paper, in markdown, or in code, they encounter questions they cannot answer independently: Did I violate the Single Responsibility Principle by combining fee calculation and ticket persistence?* Is my `Vehicle` class tightly coupled to `ParkingSpot` ?* What happens to my class hierarchy if an electric vehicle charging station is introduced tomorrow?* Did I add design patterns (e.g., Abstract Factory) unnecessarily when a simple factory was sufficient?*

2.4 Lack of an Iterative Delta Loop

Mastery requires feedback followed by immediate re-application. In existing workflows, learners rarely attempt a problem a second time because they have no mechanism to measure whether **Attempt 2** successfully addressed the structural flaws of **Attempt 1**.

3. Analysis of Existing Approaches & Market Tools

We benchmarked existing platforms and learning tools across four critical dimensions:

Solution CategoryExamplesPractice ModelFeedback MechanismCritical Gaps :

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DSA PlatformsLeetCode, HackerRankCode submission against unit test casesBinary: Pass / Fail / Timeout / MemoryTests runtime behavior and time complexity only. Completely blind to coupling, cohesion, SOLID, and class responsibilities. **Passive Content Platforms**Educative (Grokking OOD), YouTube channelsStatic reading of reference solutions and UML diagramsNone (Self-assessment)Learner cannot submit their own design. Encourages memorization of static solutions

rather than problem decomposition. **Peer / Mock Platforms**Pramp, Interviewing.io1-on-1 human mock interviewsSubjective peer or interviewer notesHigh cost, scheduling overhead, and highly variable evaluator quality. Unusable for daily, iterative self-practice. **Generic LLMs**ChatGPT, Claude (Unstructured)Freeform text prompt: "Review my parking lot design"Conversational prose and uncalibrated scores ("Looks good! 8/10")Hallucinatory ratings, moving standards across turns, lack of structured rubrics, and inability to track versioned attempt history.

Key Insight on LLMs in Design Evaluation

Large Language Models exhibit deep reasoning over code and architecture, but **unconstrained LLM prompts produce deceptive evaluations**. When prompted with "Is this a good design?", an LLM tends to be sycophantic, praises superficial naming conventions, and invents random numerical scores. To be genuinely useful, the LLM must be constrained to a **deterministic, evidence-backed rubric** where every critique cites specific lines/classes from the learner's submission.

4. Key Gaps Identified

From our comparative analysis, we identified three critical unmet needs:

- 1. The Need for Objective, Rubric-Grounded Dimensions:** Evaluation must not be a single monolithic score. It must dissect a design across orthogonal dimensions: *Domain Modeling & Requirement Coverage, Single Responsibility & Cohesion, Coupling & Extensibility, and Trade-off Awareness*.
- 2. The Need for Concrete Evidence Citations:** Vague feedback like "Improve your coupling" is useless. Effective feedback must point directly to candidate artifacts: "Concern: Your *ParkingLot* class directly instantiates *FourWheelerSpot*, creating tight coupling. Suggestion: Introduce a *SpotAllocationStrategy* interface."
- 3. The Need for Attempt Versioning:** A platform must record attempt iterations ($A_1 \rightarrow A_2 \rightarrow A_3$) to track rubric score deltas, showing the learner where they gained ground.

5. Our Product Direction: A Focused Practice Platform

Rather than attempting to build an all-in-one learning management system or a heavy cloud collaboration tool, our product solves **one clear learner problem well**:

> "Give software engineers an active, focused environment to draft Low-Level Designs, receive **deterministic and rubric-grounded semantic critiques with concrete evidence citations**, and **iterate across versioned attempts**."

THE LEARNER JOURNEY | | | 1. Problem Catalog → 2. Structured Workspace → 3. Hybrid Engine | | (Curated LLD with (Entities, Interfaces, (Deterministic | | explicit constraints) Patterns, Trade-offs) + Semantic) | | | | ▼ | | 6. Progressive Retry ← 5. Attempt History ← 4. Evidence Feedback | (Delta score tracking (Audit trail of all (Rubric scores, | | against previous try) attempts & evolution) concerns, fixes) |

Core Product Principles:

1. **Curated, Constraint-Rich Problems:** 5 representative, high-frequency LLD problems (Multi-Floor Parking Lot, Multi-Car Elevator Dispatcher, Splitwise Expense Sharing, High-Throughput In-Memory Cache with LRU/LFU/FIFO, and Distributed API Rate Limiter with Token Bucket/Leaky Bucket/Sliding Window) with explicit operational constraints, forcing learners to evaluate trade-offs rather than memorizing static templates.

2. **Structured Submission Model:** Requiring explicit articulation of (a) Core Domain Entities & Responsibilities (SRP), (b) Interfaces & Contracts (DIP/OCF), (c) Design Patterns used with justification, and (d) Concurrency Invariants & Key Trade-offs.

3. **Interactive Visual & Implementation Bridge:** Live UML Class Diagram*: Dynamically synthesizes the learner's declared entities, contracts, and pattern bindings into visual class-node cards and standard Mermaid.js syntax in real time. Multi-Language Code Boilerplate Generator*: Generates production-ready, strongly typed starter code in **Java**, **TypeScript**, and **C++**, bridging the cognitive gap between architectural abstraction and concrete code execution.

4. **Hybrid Evaluation Architecture:** Deterministic Phase*: Immediate (<50ms) validation of schema completeness, required entity counts, interface contracts, and structural integrity. Semantic Phase*: Structured rubric scoring where every score is grounded in **Concrete Evidence**, **Specific Design Smells/Concerns**, and **Actionable Remediation Suggestions**.

5. **Stateful, Resilient Attempt Lifecycle:** Explicit state transitions (SUBMITTED $\rightarrow$ EVALUATING $\rightarrow$ COMPLETED / FAILED) ensuring learner submissions are never lost, idempotent against duplicate requests, and easily retried.

6. **Evolution-Ready Seams:** Clean domain boundaries allowing the addition of new submission formats (e.g., class diagrams, code files) and new evaluator engines (e.g., static linters, peer review) without modifying the core practice orchestration.